Dynamic Route Optimization in a Mobile Museum Guide Using Sensor-Based User Interest Detection

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***Abstract*—Mobile Museum guide applications often rely on user preferences collected before the visit, which may not accurately reflect actual interests once users interact with real artifacts. This limitation can reduce the effectiveness of personalized navigation.**

This research proposes a context-aware mobile museum guide that dynamically adapts to user behavior during the visit. Initially, a route is generated based on a short user survey. To overcome the limitations of static preferences, the system incorporates a real-time interest detection mechanism. A User Interest Artifact Detector is developed using smartphone sensors, including accelerometer and gyroscope data, along with BLE beacon-based proximity tracking. User interest is inferred when the user is standing near an artifact within a defined distance, distinguishing active engagement from movement or rest.

As new interests are identified, the system updates the navigation route in real time. This adaptive approach enables more accurate personalization and enhances user engagement, demonstrating the potential of combining sensor-based activity recognition with location-aware technologies in museum environments.

Keywords—Mobile Museum Guide, User Interest Detection, BLE Beacons, Activity Recognition, Personalization

I. Introduction (*Heading 1*)

Museums have increasingly adopted digital technologies to enhance visitor engagement and improve the overall experience. Mobile museum guide applications, in particular, have become valuable tools for providing navigation support, contextual information, and personalized recommendations. These systems can help visitors explore exhibitions more effectively and engage more deeply with museum content.

However, visitor behavior often reveals inefficiencies in how museum spaces are explored. Many visitors, especially first-time visitors, enter without prior knowledge of the exhibits and tend to spend most of their time in the initial sections. As the visit progresses, fatigue and reduced attention frequently lead to decreased engagement, causing later exhibits to be overlooked. This results in an incomplete museum experience.

This issue is more pronounced among foreign visitors, who may lack familiarity with the cultural or historical context of the exhibits. Although existing systems attempt personalization through initial preference surveys, these approaches rely solely on pre-visit input and may not accurately reflect users' actual interests during the visit.

To address these limitations, this study proposes a context-aware mobile museum guide that combines initial user profiling with real-time behavior analysis. Users complete a brief survey at the start, and the system generates an initial navigation route. As the visit progresses, the system continuously refines recommendations based on observed user behavior.

The proposed system includes a User Interest Artifact Detector that uses smartphone sensors and BLE beacons to identify meaningful interactions with artifacts. By detecting patterns such as sustained proximity to exhibits, the system can distinguish genuine interest from casual movement.

Using these insights, the system dynamically updates navigation routes to reflect evolving user preferences. This approach supports a more balanced, personalized, and engaging museum experience.

The main contributions of this work are identifying inefficiencies in visitor behavior, introducing a hybrid personalization approach, and developing a dynamic navigation system that adapts in real time.

II. Related Work

Mobile museum guide systems and personalized recommendation approaches have been widely explored, with early work focusing primarily on user profiling and adaptive content delivery. One of the earliest systems, the HIPS mobile museum guide, introduced the idea of adapting tour suggestions based on user interests and location. It emphasized minimizing user distraction and providing context-aware information during the visit. However, interest modeling in such systems was largely based on navigation patterns and user interactions, without the ability to capture real-time engagement with specific artifacts.

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Similarly, the CHIP (Cultural Heritage Information Personalization) framework proposed a structured approach to personalized museum experiences through semantic enrichment and user profiling. By collecting explicit user preferences and ratings, the system generated customized tours and recommendations. While this approach improved explainability and addressed cold-start issues, it relied heavily on explicit feedback, which can be intrusive and impractical in short museum visits. Moreover, it assumed that user preferences remain relatively stable throughout the visit, limiting its ability to adapt to changing interests.

More recent research has explored adaptive and dynamic recommendation techniques. For instance, the CHESS project introduced personalized storytelling experiences by combining initial user profiles with implicit behavioral signals such as content skipping and interaction patterns. Although this approach attempted to refine user models during the visit, it still depended mainly on application-level interactions rather than direct observation of user engagement with physical exhibits. Additionally, limitations in precise indoor localization restricted the effectiveness of behavior-based adaptation.

In parallel, advances in indoor positioning technologies, particularly BLE beacon-based systems, have enabled location-aware museum applications. BLE-based approaches allow systems to estimate user proximity to artifacts and deliver context-specific content. However, these systems are often affected by signal instability, environmental interference, and limited accuracy, especially in crowded museum environments. As a result, many implementations rely on coarse location estimates and do not incorporate higher-level behavioral interpretation, such as distinguishing between passive presence and active interest.

Overall, existing systems demonstrate significant progress in personalization and navigation but share several common limitations. Most rely on static or semi-static user preferences collected before or during the visit, and few systems effectively capture real-time user interest based on in-situ behavior. Additionally, implicit signals such as dwell time or interaction patterns are often ambiguous and insufficient for accurately identifying user engagement. These limitations highlight the need for a more robust, behavior-driven approach that integrates real-time sensing with adaptive navigation.

III. Methodology

A. System Overview

The proposed system is a personalized mobile museum guide that combines survey-based user profiling, real-time behavior detection, indoor localization, and adaptive route generation. Its purpose is to improve museum navigation by identifying not only what visitors initially claim to prefer, but also what they appear to be genuinely interested in during the visit.

The system operates in two stages. In the first stage, an initial visitor profile is created from survey responses. This profile is used to generate a temporary route through the museum based on selected artifact categories. In the second stage, the system continuously observes user behavior through smartphone sensor data and indoor positioning signals. If the observed behavior indicates interest in artifacts outside the initial route, the system updates the visitor's preference model and regenerates the route accordingly.

Let the overall system state at time $t$be defined as

$$S_t = \{P_t, A_t, L_t, I_t, R_t\}, \qquad (1)$$

where P_t denotes the user profile, A_t the detected activity state, $L_t$the estimated indoor location, $I_t$the inferred interest state, and $R_t$the active navigation route. Route adaptation is then expressed as

$$R_{t+1} = f(P_t, I_t, L_t), \qquad (2)$$

where $f(\cdot)$represents the route update function based on the current profile, inferred interests, and visitor position.

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B. Survey-Based User Profiling

At the beginning of the museum visit, each visitor completes a short survey within the mobile application. The survey records two forms of information. First, it identifies the visitor type, such as student, teacher, professor, civilian, or foreign visitor. Second, it captures initial artifact preferences by asking the visitor to select categories of interest.

This stage is important because many museum visitors arrive with limited prior knowledge of the available exhibits. In the observed setting, a large proportion of visitors spend a disproportionate amount of time in the first few rooms simply because they do not know what lies ahead. As a result, later sections of the museum may be visited with reduced attention, and important artifacts may be missed. This issue is particularly relevant for foreign visitors, who may be unfamiliar with the cultural meaning of certain objects. The survey therefore serves as an initial mechanism for narrowing the content space and generating a more relevant starting route.

Let the initial profile be denoted as

$$P_0 = \{u, C_0\}, \qquad (3)$$

where $u$is the visitor type and $C_0$is the set of initially selected artifact categories. Although this profile is useful for initialization, it is treated as provisional because it is created before the visitor encounters the actual exhibits.

C. Activity Recognition for User Behavior Detection

A central component of the proposed system is the behavior detection module, which identifies whether the visitor is walking, standing, or sitting. This module is based on smartphone inertial sensing and uses data acquired from the accelerometer and gyroscope. The purpose of this step is to distinguish movement between exhibits from moments of possible engagement and from non-engagement states such as resting.

At each time step t, the sensor input vector is defined as

$$\mathbf{x}_t = [a_x(t), a_y(t), a_z(t), g_x(t), g_y(t), g_z(t)], \qquad (4)$$

where $a_x(t)$, $a_y(t)$, and $a_z(t)$are the accelerometer readings along the three axes in meters per second squared, and $g_x(t)$, $g_y(t)$, and $g_z(t)$are the gyroscope readings in radians per second.

The continuous sensor stream is segmented into fixed-length windows,

$$W_k = \{\mathbf{x}_{t_1}, \mathbf{x}_{t_2}, \dots, \mathbf{x}_{t_n}\}, \qquad (5)$$

where W_k denotes the k-th analysis window containing n time samples. From each window, descriptive features are extracted to represent motion characteristics. These may include measures such as the mean, variance, and signal magnitude over each axis. The window is then passed to a trained classification model that predicts the activity label,

$$\hat{A}_k \in \{\text{walking,standing,sitting}\}. \qquad (6)$$

In the proposed museum setting, the three activity classes are interpreted as follows. Walking indicates movement between exhibits; standing indicates possible attention toward a nearby artifact; and sitting is treated as a rest state and is therefore excluded from interest estimation. This separation is necessary because stopping near an artifact is not sufficient on its own to indicate interest unless the stop also satisfies additional spatial and temporal conditions.

D. Artifact-Level Interest Inference

Visitor interest is inferred by combining three signals: activity state, proximity to an artifact, and dwell time. The intention is to reduce the ambiguity that arises when dwell time alone is used as a proxy for attention.

Let $d_j(t)$ denote the estimated distance in meters between the visitor and artifact j at time t. An artifact is considered a candidate for interest only when the visitor is near it and is not in motion. This condition is written as

$$\hat{A}_k = \text{standing}, d_j(t) \le d_{\text{th}} \qquad (7)$$

where d_{th} is the proximity threshold. In the current design, the system focuses on close-range interaction and uses a threshold in the range of 0.0 to 0.9 meters, following your implementation concept.

In addition to proximity and posture, the visitor must remain in that valid zone for a minimum dwell time. Let T_j denote the accumulated time spent near artifact j while satisfying the standing condition. Interest is validated only if

$$T_j \ge T_{\min}, \qquad (8)$$

where $T_{\min}$ is the minimum dwell time threshold in seconds. The exact threshold may be tuned experimentally during implementation.

$$I_j = \begin{cases} 1, & \text{if } \hat{A}_k = \text{standing}, d_j(t) \le d_{\text{th}}, \text{ and } T_j \ge T_{\min}, \\ 0, & \text{otherwise.} \end{cases} \qquad (9)$$

This formulation intentionally excludes sitting from interest detection. In a museum environment, sitting is more plausibly associated with resting, waiting, or disengagement, and therefore it is not counted as evidence of attention toward an artifact.

E. Enhanced Interest Scoring

To improve robustness, the system does not rely solely on a binary decision. Instead, an enhanced interest score is computed for each artifact by combining multiple indicators. This reflects the fact that interest is often gradual rather than purely discrete.

Let T_j represent valid dwell time for artifact j, V_j the number of revisits to artifact j, and Q_j a confidence value derived from behavior and localization reliability. The artifact-level interest score is then defined as

$$\text{Score}_j = \alpha T_j + \beta V_j + \gamma Q_j, \qquad (10)$$

where α, β, and γ are weighting coefficients. These coefficients may be normalized or tuned empirically during system calibration.

This enhanced formulation is preferable to a simple standing-time count because a visitor may briefly stop near an object for reasons unrelated to interest, such as congestion or navigation hesitation. By incorporating revisit behavior and confidence information, the model becomes less sensitive to accidental pauses and noisy location estimates.

F. Category Preference Modeling

Each artifact in the museum is associated with a category, such as clothes, coins, tools, jewelry, ceramics, sculptures, or manuscripts. Once artifact-level interest scores are computed, the system aggregates them at the category level to estimate the visitor's evolving preferences.

Let C_m denote the m-th artifact category and let $a_j \in C_m$ indicate that artifact j belongs to category C_m. The cumulative category preference score is computed as

$$\text{Pref}(C_m) = \sum_{a_j \in C_m} \text{Score}_j. \qquad (11)$$

The resulting category scores are ranked in descending order, and the highest-ranked categories are treated as the visitor's current priorities. This representation allows the system to move beyond single-artifact reactions and instead learn broader thematic interests. Such category-level modeling is especially useful in museums because visitors often display interest in related groups of objects rather than in isolated exhibits.

G. Dynamic Route Generation

The initial route is generated from the survey-based profile, but the final navigation path is continuously updated as the visitor's observed preferences evolve. This dynamic behavior is one of the key contributions of the proposed system.

The museum is represented as a graph

$$G = (N, E), \qquad (12)$$

where N is the set of museum nodes, such as rooms, corridors, or exhibit zones, and E is the set of walkable connections between nodes. Each artifact or artifact category is mapped to one or more nodes in this graph.

Let the navigation route at time t be represented by

$$R_t = \{n_1, n_2, \dots, n_k\}, \qquad (13)$$

where $n_i \in N$ denotes a route node. The goal is to generate a route that simultaneously preserves path efficiency and prioritizes high-interest categories.

To formalize this, a route cost function is defined as

$$J(R) = \sum_{i=1}^{k-1} d(n_i, n_{i+1}) + \lambda \sum_{i=1}^{k} \phi(n_i), \qquad (14)$$

where $d(n_i, n_{i+1})$is the walking distance in meters between consecutive nodes, $\phi(n_i)$ is a penalty term that decreases when node $n_i$leads to a preferred category, and λcontrols the balance between route efficiency and personalization.
The updated route is then selected as

$$R_{t+1} = \arg\min_{R \in \mathcal{R}} J(R), \qquad (15)$$

where $\mathcal{R}$is the set of feasible routes from the visitor's current location to the remaining relevant museum sections. This formulation allows the system to introduce newly discovered interests into the route while still maintaining a coherent path through the museum.

H. BLE-Based Indoor Localization

Accurate identification of the nearest artifact requires a reliable estimate of the visitor's indoor location. For this purpose, the proposed system uses Bluetooth Low Energy beacons distributed throughout the museum. The mobile device scans nearby beacons and records their Received Signal Strength Indicator values in decibels relative to one milliwatt.

1) RSSI Smoothing

Because raw RSSI values are highly unstable due to obstacles, body blocking, and multipath effects, the signals are first smoothed using an exponential moving average,

$$\tilde{r}_t = \eta r_t + (1-\eta)\tilde{r}_{t-1}, \qquad (16)$$

where r_t is the raw RSSI measurement at time t, $\tilde{r}_t$ is the smoothed RSSI value, and η is the smoothing factor. In practice, η may be selected in the range 0.20 to 0.35, depending on the desired balance between responsiveness and stability.

2) RSSI-to-Distance Conversion

The smoothed RSSI value is converted into an estimated distance using the log-distance path-loss model,

$$d_i = 10^{\frac{r_0 - \tilde{r}_i}{10n}}, \qquad (17)$$

where $d_i$is the estimated distance in meters from beacon i, $r_0$is the calibrated RSSI value at one meter, and $n$is the path-loss exponent determined by the indoor environment.

3) Weighted Centroid Localization

Instead of using strict trilateration, the proposed system employs a weighted centroid approach because it is computationally lighter and more stable under fluctuating RSSI conditions. Each beacon contributes to the position estimate according to a weight inversely proportional to its distance,

$$w_i = \frac{1}{d_i}. \qquad (18)$$

If the selected beacons have coordinates (x_i, y_i) , the estimated visitor position $(\hat{x}, \hat{y})$is computed as

$$\hat{x} = \frac{\sum_{i=1}^{M} w_i x_i}{\sum_{i=1}^{M} w_i}, \hat{y} = \frac{\sum_{i=1}^{M} w_i y_i}{\sum_{i=1}^{M} w_i}, \qquad (19)$$

where M is the number of selected beacons. In implementation, only the strongest 3 to 4 beacon signals are retained in order to reduce the effect of distant and unstable measurements.

4) Position Smoothing

To improve visual continuity and prevent abrupt marker movement on the map, temporal smoothing is applied to the estimated position,

$$\mathbf{p}_t = \rho \hat{\mathbf{p}}_t + (1-\rho)\mathbf{p}_{t-1}, \qquad (20)$$

where $\hat{\mathbf{p}}_t$is the raw estimated position, $\mathbf{p}_{t-1}$is the previous smoothed position, and ρ is the position smoothing coefficient.

5) Outlier Rejection and Map Matching

If the implied movement speed between two consecutive positions exceeds a plausible indoor walking limit, the update is rejected as an outlier. After this filtering step, the position is projected onto the nearest valid corridor or path segment in the museum map to ensure that the displayed location remains consistent with navigable space. This final step improves both usability and route stability.

I. Integrated System Workflow

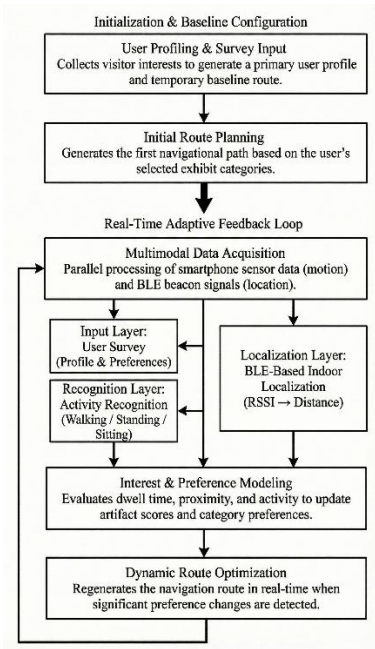


Figure 1: Architecture of the proposed adaptive museum guidance system with real-time behavior detection and dynamic route optimization.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The proposed system was implemented as a mobile application and evaluated in an indoor museum-like environment. The setup consisted of a smartphone equipped

with accelerometer and gyroscope sensors for activity recognition, along with a network of Bluetooth Low Energy (BLE) beacons deployed across multiple exhibit areas. The beacons were placed at key artifact locations to enable proximity-based localization.

Sensor data were continuously collected during user movement and segmented into fixed time windows for activity classification. The activity recognition model was trained to classify three primary states: walking, standing, and sitting. For interest detection, a minimum dwell time threshold of 12 seconds was used, along with a proximity constraint of 0.9 meters from the artifact.

B. Activity Recognition Performance

The activity recognition module demonstrated stable performance across the three defined classes. Walking and standing were clearly distinguishable due to differences in motion patterns, while sitting was identified through low variance in both accelerometer and gyroscope signals.

The overall classification accuracy reached approximately 93.4%, with standing detection achieving the highest reliability. This level of performance was sufficient for downstream interest inference, as the system primarily relies on accurate identification of stationary (standing) behavior near artifacts.

C. BLE Localization Performance

The BLE-based indoor localization module provided reasonably accurate proximity estimation for artifact detection. After applying signal smoothing and weighted centroid localization, the average positioning error was observed to be within 0.6-1.2 meters, depending on environmental conditions such as crowd density and signal interference.

Although RSSI-based localization is inherently noisy, the use of exponential smoothing and beacon selection (top 3 - 4 strongest signals) significantly improved stability. The system was able to reliably identify the nearest artifact zone, which is sufficient for interest detection even without precise coordinate-level accuracy.

D. Interest Detection Results

The effectiveness of the proposed interest inference mechanism was evaluated by analyzing user behavior near selected artifacts. Table I presents representative results demonstrating how activity state, distance, and dwell time contribute to interest detection.

TABLE I: Interest Detection Results

Artifact	Activity	Distance	Dwell Time	Interest Detected
Coins	Standing	0.70m	12.3s	Yes
Cloths	Siting	0.63m	55.7s	No
Statue	Standing	0.55m	6.0s	No
Arts	Walking	-	1.2s	No

The results show that interest is correctly identified only when all conditions are satisfied. In particular, standing behavior combined with close proximity and sufficient dwell time proved to be a reliable indicator of user engagement. Short stops or movements near artifacts did not trigger false positives, indicating that the multi-condition approach effectively filters out noise.

E. Preference Modeling and Adaptation

Based on accumulated interaction data, the system computes category-level preferences. Figure 2 illustrates an example distribution of user interest across artifact categories, derived from dwell time and interaction patterns.

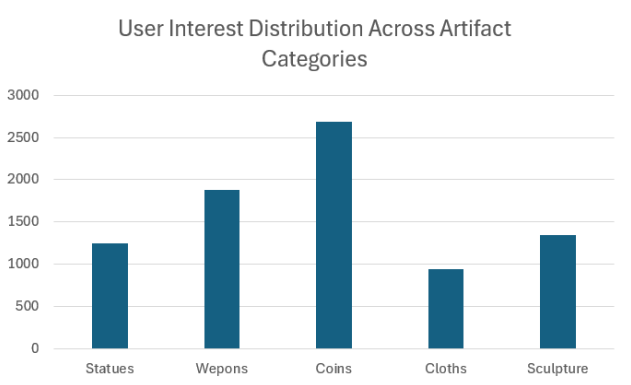


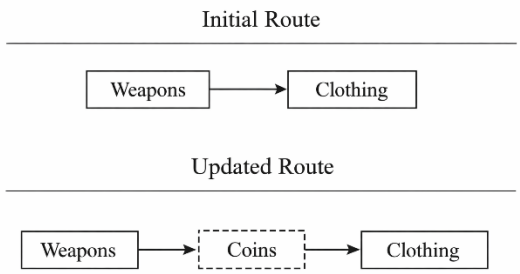
Figure 2: Category-Level User Interest Based on Total Dwell Time

In this example, the user initially selected weapon-related artifacts during the survey phase. However, during the visit, the system observed longer dwell times near coin-related exhibits. As a result, the preference model ranked the "Coins" category higher than initially selected categories.

This demonstrates the system's ability to capture evolving user interests rather than relying solely on pre-defined preferences.

F. Dynamic Route Adaptation

The impact of real-time interest detection on navigation was evaluated through scenario-based analysis. Figure 1 illustrates a representative example of route adaptation.



- Initial route: Weapons → Clothing
- Updated route: Weapons → Coins → Clothing

In this case, the system detected sustained user engagement with coin-related artifacts that were not part of the initial

route. The navigation path was then dynamically updated to include the relevant section, ensuring that the user did not miss content aligned with their actual interests.
This adaptive behavior addresses a common issue observed in traditional museum visits, where users tend to spend excessive time in early sections and lose interest before reaching later exhibits.

G. Discussion

The results indicate that combining activity recognition, BLE-based proximity detection, and dwell-time analysis provides an effective framework for real-time interest inference. Unlike conventional systems that rely on static preferences or explicit feedback, the proposed approach captures implicit behavioral signals with minimal user interaction.

One of the key strengths of the system is its ability to adapt continuously during the visit. By updating the route based on observed behavior, the system reduces the likelihood of users missing relevant exhibits and improves overall engagement.

However, several limitations remain. BLE-based localization is sensitive to environmental factors such as signal interference and human movement, which can affect distance estimation accuracy. Additionally, activity recognition performance may vary depending on device placement and user behavior patterns. The selection of dwell time thresholds also influences sensitivity and may require further tuning for different museum settings.

Despite these challenges, the proposed system demonstrates a practical and scalable approach to personalized museum guidance. By integrating sensor-based behavior detection with adaptive routing, it provides a more responsive and user-centered experience compared to traditional static systems.

V. Conclusion

This research presented a personalized mobile museum guide system that integrates user profiling, sensor-based behavior detection, and BLE-based indoor localization to enable adaptive navigation. The system addresses a key limitation of conventional museum guide applications, which rely heavily on static user preferences collected prior to the visit. By incorporating real-time user behavior, the proposed approach is able to dynamically identify visitor interests and adjust navigation routes accordingly.

The experimental results demonstrated that the combination of activity recognition, proximity estimation, and dwell time analysis provides a reliable mechanism for detecting user engagement with artifacts. The dynamic route adaptation further enhances the visitor experience by ensuring that relevant exhibits are included even when they were not part of the initial selection. This helps reduce the common issue of visitors spending excessive time in early sections while missing important artifacts in later areas.

Despite these advantages, the system is subject to certain limitations, particularly the sensitivity of BLE-based localization to environmental conditions and the dependency of activity recognition on sensor quality and device placement. Future work will focus on improving localization accuracy through sensor fusion techniques and exploring more advanced learning models for behavior and interest prediction. Additionally, incorporating contextual factors such as crowd density and exhibit popularity could further enhance route optimization and user experience

References

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